- Lab report -
Lab 03:
Electromyography Sensor

[S26] Sensors and Sensing

- Innopolis University -

Anas Khmais Hamrouni
Vladimir Toporkov
Andrey Krasnov
Askar Kadyrgulov

B23-RO-01

[Lab notebook]

[Lab Solution Git Repository]:

**experiments steps/instructions in README*

Topic :

Electromyography (EMG) Signal Acquisition and Motor Control Using Surface EMG Sensor with Arduino Uno.

Purpose:

- Learning how to correctly attach surface EMG electrodes to a muscle and selecting optimal placement for maximum signal response.
- Calibrating the EMG signal by removing baseline drift and applying signal filtering.
- Integrating the EMG signal into a closed-loop motor position control system capable of rotating a motor to 90° in one or two directions based on muscle activation.

Description of the sensor application

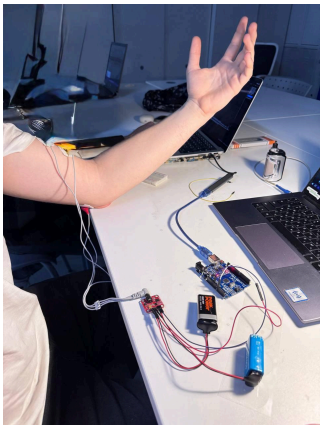
Explains how a bipolar differential surface EMG amplifier works, differential electrode placement, common-mode noise rejection, analog output, digital filtering pipeline, and how this forms the basis of myoelectric prosthetic/exoskeleton control.

List of used equipment/stack:

- **Hardware:** (documentation & datasheets in lab notebook)
  - EMG Amplifier module (surface EMG amplifier board with 3-lead electrode cable: 2 active + 1 reference electrode)
  - Disposable Ag/AgCl surface electrodes (self-adhesive, for electrode-skin contact)
  - 2 × 9 V Crone batteries (providing bipolar ±9 V supply to the EMG amplifier)
  - Arduino Uno board for serial signal acquisition
  - BLDC Motor MF4005 V2 (brushless DC motor used as actuator for position control)
  - CANable CAN <-> micro USB adapter (for CAN protocol communication with the MF4005 motor)
  - USB cables, wires, breadboard
  - PC with Python (Scipy, Matplotlib, Numpy, Pyserial) for signal visualization and serial communication

Description of experiments:

Experiment 1 — Electrode Placement Optimization



Surface EMG electrodes were tested at multiple forearm/upper-arm locations to maximize separability between relaxed and contracted muscle states. The two active electrodes were placed close together (about 1–2 cm), aligned with muscle fibers, and the reference electrode was placed on a low-activity bony area. Signal quality was evaluated from live serial data by comparing noise level at rest versus peak amplitude during voluntary squeeze. The placement with the clearest rest/contract contrast was selected for control experiments.

Experiment 2 — Signal Calibration and Envelope Extraction

The acquisition firmware performs a phase-based protocol and onboard preprocessing:
1. Timed protocol: RELAX1 (5 s) → PREPARE (3 s) → SQUEEZE (5 s) → RELAX2 (5 s).
2. Sampling: Analog EMG is sampled every 5 ms (≈200 Hz).
3. Baseline calibration: During RELAX1, the resting ADC mean is accumulated; at PREPARE, this mean is fixed as baseline.
4. Centering: AC component is computed as ac = raw − baseline.
5. Rectification: rect = |ac|.
6. Envelope: A moving-average window of 20 samples is applied (≈100 ms smoothing at 200 Hz).
Processed values (raw, ac, rect, envelope) are streamed as CSV for visualization and control.

Experiment 3 — EMG-Driven Single-Direction Motor Control with Return-to-Home

The motor controller uses the EMG envelope as a continuous effort signal rather than a fixed-angle trigger:
Mode switching with hysteresis: Enter active mode when filtered_envelope > EMG_ON. Return mode when filtered_envelope < EMG_OFF.
Effort-to-speed mapping (active mode): Envelope is normalized and mapped to velocity using a nonlinear gain (square-root), so stronger contractions produce higher target velocity.
Trajectory generation: Desired position is integrated from velocity and limited to a safe one-direction range from initial position (q0 to (q0 + XMAX_DEG)).
Relax behavior: When muscle activity drops, the controller commands a smooth return toward q0 with speed/acceleration limits.
Low-level actuation: A PD law converts position/speed error to motor current command over CAN, with current clamping for safety.
Logging: Time, motor state, current command, and EMG features are stored for post-analysis.

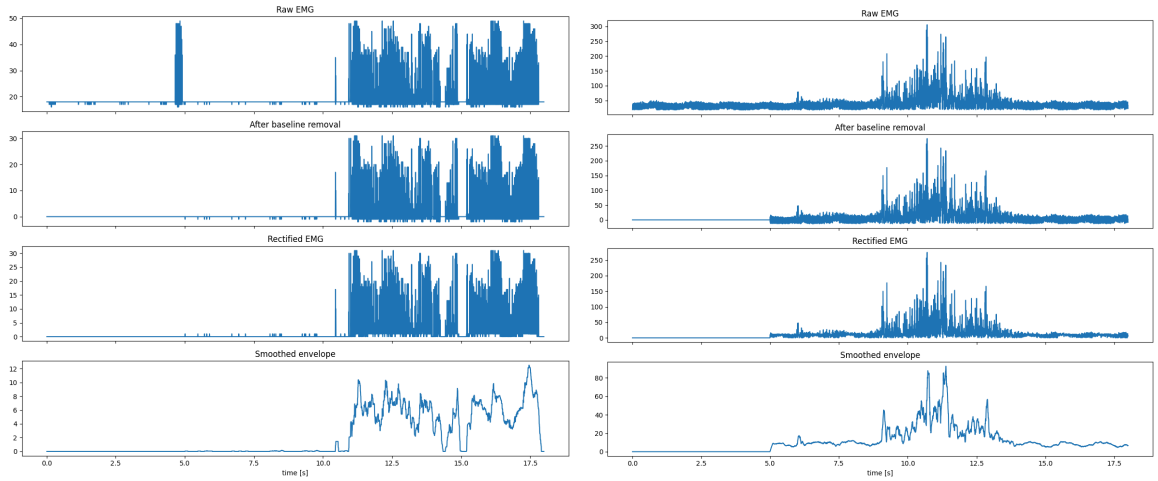
Experiment 4 — Offline Replay and Parameter Tuning from Recorded CSV

To tune control without requiring live EMG acquisition, recorded EMG CSV logs were replayed through the same control logic. This enabled repeatable adjustment of thresholds and mapping parameters (EMG_ON, EMG_OFF, EMG_MAP_MAX) while observing motor response and logged outputs. The replay mode was used to diagnose issues such as speed saturation (constant motor speed despite higher effort) and to refine sensitivity versus stability.

Description of the results of the experiments:

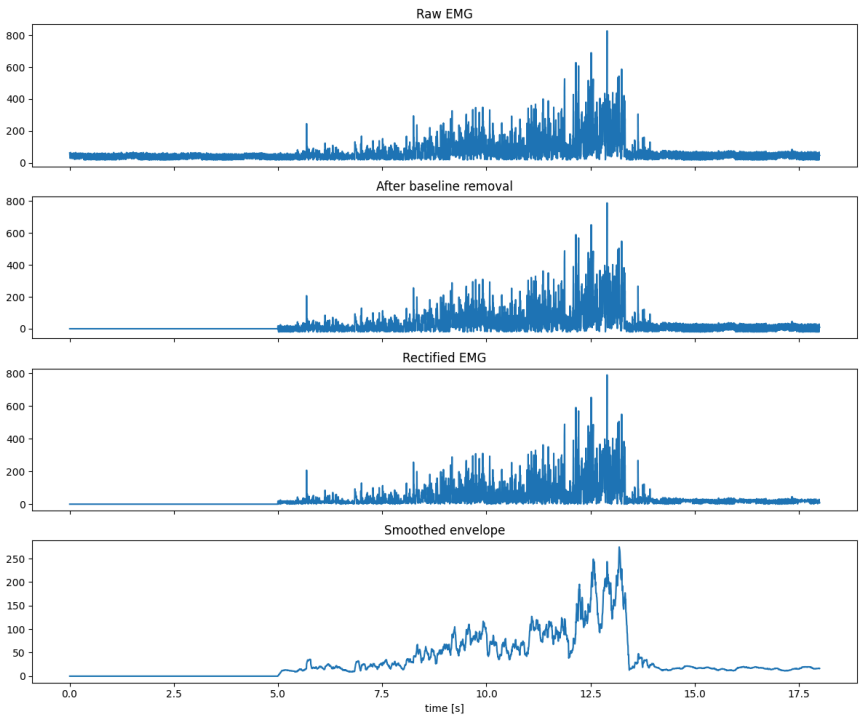
Experiment 1 — Electrode Placement Optimization

The tested placements produced clearly different signal quality. The selected placement gave the largest separation between rest and contraction: during rest, the signal stayed in a narrow low-variance band, while voluntary squeeze produced pronounced bursts with much higher amplitude. This improved the signal-to-noise ratio and made threshold-based activation reliable, so the selected placement was used for all subsequent tests.



Experiment 2 — Signal Calibration and Filtering

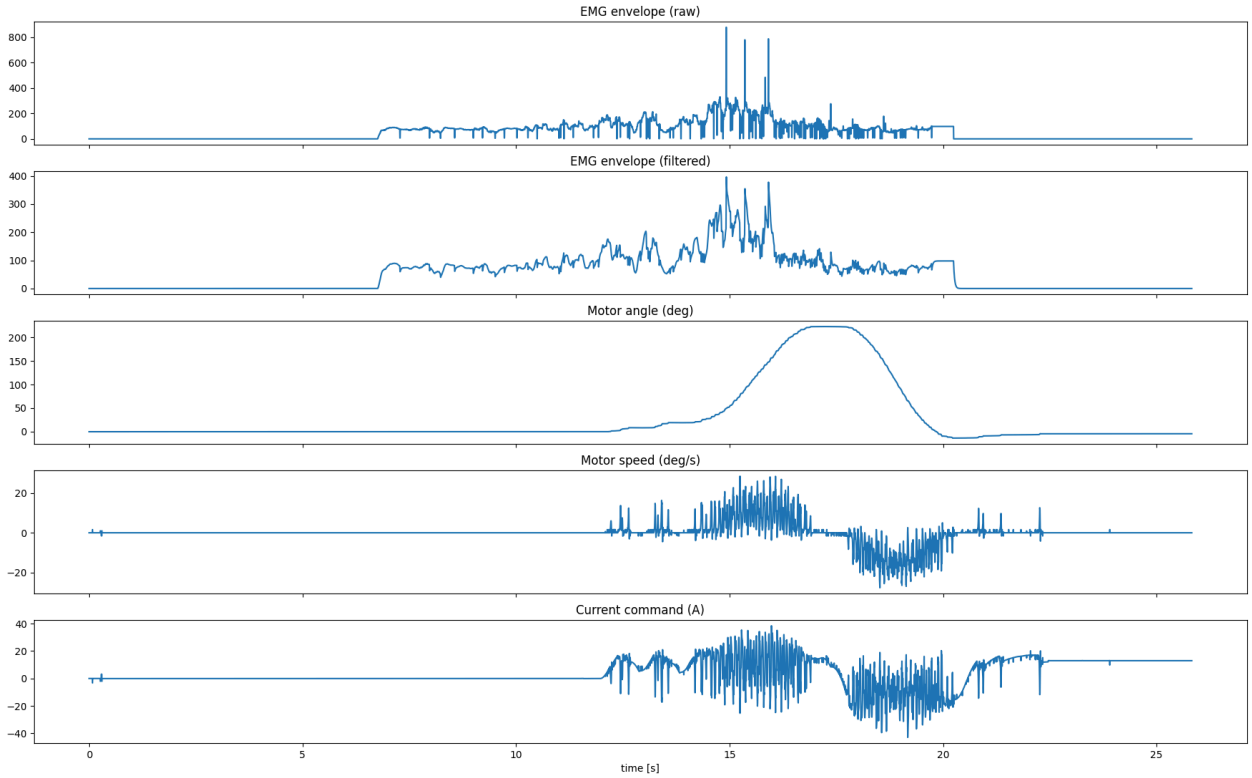
The preprocessing chain successfully converted noisy raw EMG into a usable control signal. Baseline subtraction centered the signal around zero, rectification preserved muscle effort magnitude, and moving-average smoothing produced a stable envelope. In the plots, the envelope remains low during relax phases and rises strongly during squeeze phases (with clear peaks), confirming that the processing pipeline captures muscle activation while suppressing high-frequency fluctuations. This made it possible to define robust ON/OFF control thresholds with hysteresis.



Experiment 3 — EMG-Driven Single-Direction Motor Control (D-link)

Motor behavior followed muscle effort as expected:
- When the filtered envelope crossed the activation threshold, motor angle increased in one direction.
- Stronger contractions produced higher motor speed (up to the configured limit), while weak activity produced slower motion.
- When activity dropped below the OFF threshold, the controller switched to return mode and the motor moved back toward home.
- Position and current remained bounded by software limits, and no unstable oscillation was observed in normal operation.

From the recorded run, the motor angle rose to roughly 220° during active contraction and then returned close to home after relaxation, while speed changed sign appropriately between forward and return phases. This confirms correct mode switching and trajectory generation.



Experiment 4 — Offline Replay and Parameter Tuning from CSV

Replay mode reproduced the same control logic without requiring live serial acquisition and enabled repeatable tuning. Analysis of logged envelope distributions showed a strong gap between relax and active states, allowing cleaner threshold selection. After tuning, false activations during rest were reduced and effort-dependent speed changes became more noticeable. The replay workflow also helped diagnose saturation cases (constant speed despite stronger effort) and adjust mapping parameters accordingly.